

Cuckoo: A Malware Analyser Companies Can Depend Upon

Cuckoo is a leading open source malware analyser. The Cuckoo sandbox facilitates effective analysis of various types of malware by monitoring their behaviour in a secure and isolated environment with the help of virtual machines. Based on Python, it is definitely a must-have tool in the armoury of any computer security expert.



'M'alware' is a term that refers to all types of malicious software, and is a serious threat to the IT backbone of any organisation. The impact of malware is far-reaching in both technical and economical dimensions. The wide spectrum of malware, comprising various types of attacks targeting different vulnerabilities, have made their detection and analysis a complex mission. Figure 1 illustrates a few important types of malware.

Though the mode of attack, payload, etc., of one type of malware might be different from that of another category, all of them hamper the smooth functioning of a system. Hence, it becomes mandatory to restrict their proliferation and protect systems from their negative impact. To achieve this goal, one important step is to analyse the malware's various dimensions. This article explores a leading open source

malware analysis tool called Cuckoo.

Malware analysis

Malware analysis is the task of putting a malware under the scanner, and checking its structure and behaviour. This analysis is classified into two types, as follows:

- **Static analysis:** This analysis focuses on the different components of the malware file, and is primarily based on the structure of these files. It uses the principles of reverse engineering.
- **Dynamic analysis:** This works by logging the behaviour of the malware in a safeguarded environment, like a sandbox. Based on the nature of activities that the analysed malware have carried out inside the sandbox, the future sequence of actions is decided upon.